

GIHEN DHARMARATHNE

Data Science Undergraduate

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PROFESSIONAL SUMMARY

Highly motivated Data Science undergraduate at General Sir John Kotelawala Defence University with a strong foundation in sports analytics, statistical modeling, and data-driven communication. Skilled in Python and analytical frameworks to transform complex performance and raw data into actionable strategic insights. Skilled in building predictive models and advanced visualizations that bridge the gap between technical data outputs and organizational goals, ensuring that analytical findings are effectively communicated to drive success in high-performance environments.

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML, CSS

Machine Learning & AI: Computer Vision (YOLO), Natural Language Processing (NLP), TensorFlow, Predictive Modeling

Libraries & Frameworks: Streamlit, Pandas, NumPy, Scikit-Learn, OpenCV, matplotlib, seaborn

Data & Analytics: Web Scraping, Exploratory Data Analysis (EDA), Sports Analytics, Interactive Dashboards

Data Science Tools: Microsoft Power BI, Jupyter Notebook, Google Colab, R Studio

PROJECTS

Automated Wide Ball Detection System

Python, OpenCV, YOLO, Computer Vision

[GitHub Repo](#)

- Engineered a computer vision system utilizing object detection algorithms to automate the identification of wide balls in cricket match footage.
- Processed video data to track ball trajectory relative to the crease, aiming to reduce human umpiring error in real-time.

Heart Disease Prediction Model

Python, Scikit-Learn, Pandas, Machine Learning

[GitHub Repo](#)

- Developed a predictive classification model to assess heart disease risk based on clinical parameters, optimizing for high sensitivity and accuracy.
- Conducted extensive exploratory data analysis (EDA) and feature engineering to identify key health indicators.

CricMiner: Cricket Analytics Application

Python, Web Scraping, Streamlit, Statistical Modeling

[GitHub Repo](#)

- Architected a full-stack data mining application to scrape and synthesize player performance statistics from major cricket databases.
- Implemented interactive visualization modules to analyze player career trajectories and assist in scouting.

Matchup Simulator

Python, Data Simulation, Streamlit, Pandas

[GitHub Repo](#)

- Built a probabilistic simulation engine to forecast batter-vs-bowler outcomes specifically for Sri Lankan cricket players using historical data.
- Modeled various in-game scenarios to provide strategic insights regarding bowler rotation and field placement optimization.

LPL Death Overs Batting Analysis

Data Visualization, Strategic Analysis, Power BI, Python

GitHub Repo

- Analyzed Lanka Premier League (LPL) data to evaluate batting efficiency during death overs, identifying high-impact players and scoring trends.
- Produced actionable reports on strike rates and boundary frequencies to bridge the gap toward optimal run-scoring targets.

2026 T20 World Cup Analysis

Data Visualization, Strategic Analysis, Power BI, Python

GitHub Repo

- Designed a dynamic dashboard to track team and player statistics across the T20 World Cup, highlighting key performance indicators (KPIs).
- Synthesized complex tournament data into intuitive visual formats, enabling users to compare team strengths instantly.

Policy Analysis & Adaption System

Python, NLP, Pandas, Large Language Models (LLMs)

GitHub Repo

- Developed an automated system utilizing Natural Language Processing (NLP) to analyze and summarize key insights from complex policy documents.
- Implemented adaptive algorithms to dynamically track policy changes and generate data-driven compliance recommendations.

Live Cricket Score Dashboard

Python, API Integration, Streamlit

GitHub Repo

- Built an interactive dashboard that fetches and visualizes match data, focusing on run rates, wagon wheels, and partnership statistics.
- Implemented real-time data handling to provide instant updates and comparative visualizations for ongoing matches.

EPC Energy Analysis Dashboard

Power BI, DAX, Data Modeling, Data Visualization

GitHub Repo

- Developed an interactive Power BI dashboard to analyze EPC data, tracking building energy efficiency and carbon emission metrics.
- Implemented complex DAX measures and optimized data models to provide actionable visual insights for energy optimization.

Road Safety Analysis Dashboard

Power BI, SQL, Data Visualization, Data Cleaning

GitHub Repo

- Developed an interactive dashboard to analyze road accident incidents, identifying high-risk locations and casualty trends.
- Transformed raw accident data into visual reports utilizing geospatial mapping and time-series analysis to improve safety measures.

CERTIFICATIONS

Rajasthan Royals School of Business: Sports Analytics and Management Certification

EDUCATION

General Sir John Kotelawala Defence University

B.Sc. in Applied Data Science Communication

Bandaranayake College

G.C.E. Advanced Level

Ratmalana, Sri Lanka

2024 – Present

Gampaha, Sri Lanka

2020 – 2022

LEADERSHIP & ACTIVITIES

Social Media Analyst | DatAInspire_KDU

General Sir John Kotelawala Defence University

Jan 2025 – 2026

Ratmalana, Sri Lanka

- Directing digital engagement strategies for the Club, including content scheduling and real-time audience trend tracking for events and campaigns.
- Synthesized monthly performance reports for the executive committee, ensuring data-driven decision-making in content timing and platform selection.
- Collaborated with stakeholders to align promotional activities with organizational goals, maintaining consistent brand visibility to drive audience growth.

Member | University Cricket Team

General Sir John Kotelawala Defence University

2024 – 2026

Ratmalana, Sri Lanka

- Representing the university in inter-university tournaments, demonstrating teamwork, discipline, and high-pressure decision-making skills.
- Actively participating in rigorous training sessions and tactical planning to enhance team performance and maintain peak physical fitness.

Assistant Coach | MB Cricket Academy

Negombo, Sri Lanka

2023 – Present

- Mentoring young athletes on cricket fundamentals, focusing on technique development, game awareness, and sportsmanship.
- Assisting in the design and execution of specialized training drills for batting and bowling to improve individual and team performance.
- Monitoring player progress and providing constructive feedback to help students achieve their athletic goals.